# PRACTICAL 6

## Aim

To develop and execute an e-commerce sales analysis solution using Apache Spark DataFrames and PySpark to process large-scale online retail transaction data and generate useful business insights.

The practical focuses on loading the Online Retail dataset into a Spark DataFrame, examining its schema and data types, cleaning invalid transactions, calculating revenue, analyzing topselling products, performing country-wise revenue analysis, calculating average revenue per order, applying discounts, and generating monthly revenue summaries

## Problem Definition

An e-commerce organization maintains a large collection of online retail transaction records containing invoice numbers, product codes, product descriptions, quantities, invoice dates, unit prices, customer identifiers, and countries. Traditional spreadsheet-based tools may become difficult to use for processing large transaction datasets. Therefore, the organization uses Apache Spark DataFrames for scalable data processing and analysis. As a Data Analytics Engineer, the task is to process the UCI Online Retail dataset using PySpark and generate meaningful sales insights. The practical includes loading the dataset into a Spark DataFrame, displaying the dataset and schema, validating column data types, checking missing values, removing invalid transactions, creating a Revenue column using Quantity × UnitPrice, identifying top-selling products, calculating country-wise revenue, finding the top revenuegenerating countries, calculating average revenue per order, applying a discount, identifying low-performing products, and generating monthly revenue summaries. The dataset contains 541,909 transaction records before data filtering.

## Objectives

- Understand Apache Spark DataFrames and PySpark.
- Create and configure a SparkSession.
- Load the Online Retail dataset into a Spark DataFrame.
- Display and examine the dataset.
- Display the schema of the DataFrame.
- Validate the data types of all columns.
- Check for missing values.
- Remove cancelled transactions.
- Remove records with invalid quantities.

- Remove records with invalid unit prices.
- Create a Revenue column using Quantity × UnitPrice.
- Calculate total revenue.
- Calculate total quantity sold.
- Calculate the total number of orders.
- Identify top-selling products by quantity.
- Identify top products by revenue.
- Perform country-wise revenue analysis.
- Find the top three revenue-generating countries.
- Calculate average revenue per order.
- Add a discount column.
- Calculate revenue after discount.
- Identify low-performing products.
- Generate monthly revenue summaries.
- Understand Spark DataFrame transformations and aggregations.
- Interpret the results for business analysis.

# Key Questions / Analysis

**Q1. Why are Spark DataFrames preferred over traditional spreadsheets for largescale analytics?**

Spark DataFrames are designed for distributed data processing and can handle large datasets efficiently. They provide operations for filtering, grouping, aggregation and transformation while allowing computation to be distributed across available processing resources.

Traditional spreadsheets are generally more suitable for smaller datasets, whereas Spark DataFrames are designed for scalable data processing.

**Q2. What is the role of the groupBy() operation in data aggregation?**

The groupBy() operation groups records based on one or more columns.

For example:

df.groupBy("Country") groups all transactions

belonging to the same country.

Aggregation functions such as:

sum() avg() count() can then be

applied to each group.

In this practical, groupBy() was used for:

- Product analysis
- Country-wise revenue analysis
- Invoice-level revenue analysis

**Q3. Explain the difference between transformations and actions in Spark.**

Transformations

Transformations create a new DataFrame based on an existing DataFrame.

Examples:

filter()

withColumn()

groupBy()

orderBy()

Actions

Actions trigger computation and return results.

Examples:

show() count()

collect()

Spark uses lazy evaluation for transformations and performs the actual computation when an action is executed.

**Q4. How does Spark improve scalability for e-commerce analytics?**

Spark supports distributed processing, allowing large datasets to be divided into smaller processing tasks.

This provides:

1. Scalability – large datasets can be processed using additional resources. 2.

Parallel processing – multiple records can be processed simultaneously.

3. Efficient aggregation – large datasets can be grouped and summarized.
4. Fault tolerance – Spark can recover from certain processing failures.
5. Large-scale processing – millions of transaction records can be analyzed using DataFrames.

**Q5. Which category contributes the highest revenue and what business action can be taken?**

The selected UCI Online Retail dataset does not contain an explicit Category field. Therefore, category-wise revenue could not be directly calculated without making assumptions about product categories.

Instead, product-wise and country-wise revenue analysis was performed.

The highest displayed product revenue was associated with DOTCOM POSTAGE, while country-wise analysis showed the United Kingdom with the largest revenue value in the generated results.

# Supplementary Problems

1. Find the top 3 revenue-generating geographical regions.
2. Calculate average revenue per order.
3. Add a discount column and recalculate revenue.

# Output

```
from pyspark.sql import SparkSession
from pyspark.sql.functions import col, sum, avg, desc, asc, round
from pyspark.sql.functions import to_timestamp, date_format, sum, col
from pyspark.sql.functions import expr, coalesce, date_format, sum, round
```

```
spark = SparkSession.builder \
    .appName("ECommerce Sales Analysis") \
    .master("local[*]") \
    .getOrCreate()

print("Spark Session Created Successfully")
```

```
Spark Session Created Successfully
```

```
df = spark.read.csv(
    "data.csv",
    header=True,
    inferSchema=True
)
```

```
df.show()
```

```
+---------+---------+--------------------+--------+--------------+---------+----------+--------------+
|InvoiceNo|StockCode|         Description|Quantity|   InvoiceDate|UnitPrice|CustomerID|       Country|
+---------+---------+--------------------+--------+--------------+---------+----------+--------------+
|   536365|   85123A|WHITE HANGING HEA...|       6|12-01-2010 08:26|     2.55|     17850|United Kingdom|
|   536365|    71053| WHITE METAL LANTERN|       6|12-01-2010 08:26|     3.39|     17850|United Kingdom|
|   536365|   84406B|CREAM CUPID HEART...|       8|12-01-2010 08:26|     2.75|     17850|United Kingdom|
|   536365|   84029G|KNITTED UNION FLA...|       6|12-01-2010 08:26|     3.39|     17850|United Kingdom|
|   536365|   84029E|RED WOOLLY HOTTIE...|       6|12-01-2010 08:26|     3.39|     17850|United Kingdom|
|   536365|    22752|SET 7 BABUSHKA NE...|       2|12-01-2010 08:26|     7.65|     17850|United Kingdom|
|   536365|    21730|GLASS STAR FROSTE...|       6|12-01-2010 08:26|     4.25|     17850|United Kingdom|
|   536366|    22633|HAND WARMER UNION...|       6|12-01-2010 08:28|     1.85|     17850|United Kingdom|
|   536366|    22632|HAND WARMER RED P...|       6|12-01-2010 08:28|     1.85|     17850|United Kingdom|
|   536367|    84879|ASSORTED COLOUR B...|      32|12-01-2010 08:34|     1.69|     13047|United Kingdom|
|   536367|    22745|POPPY'S PLAYHOUSE...|       6|12-01-2010 08:34|      2.1|     13047|United Kingdom|
|   536367|    22748|POPPY'S PLAYHOUSE...|       6|12-01-2010 08:34|      2.1|     13047|United Kingdom|
|   536367|    22749|FELTCRAFT PRINCES...|       8|12-01-2010 08:34|     3.75|     13047|United Kingdom|
|   536367|    22310|IVORY KNITTED MUG...|       6|12-01-2010 08:34|     1.65|     13047|United Kingdom|
|   536367|    84969|BOX OF 6 ASSORTED...|       6|12-01-2010 08:34|     4.25|     13047|United Kingdom|
|   536367|    22623|BOX OF VINTAGE JI...|       3|12-01-2010 08:34|     4.95|     13047|United Kingdom|
|   536367|    22622|BOX OF VINTAGE AL...|       2|12-01-2010 08:34|     9.95|     13047|United Kingdom|
|   536367|    21754|HOME BUILDING BLO...|       3|12-01-2010 08:34|     5.95|     13047|United Kingdom|
|   536367|    21755|LOVE BUILDING BLO...|       3|12-01-2010 08:34|     5.95|     13047|United Kingdom|
|   536367|    21777|RECIPE BOX WITH M...|       4|12-01-2010 08:34|     7.95|     13047|United Kingdom|
+---------+---------+--------------------+--------+--------------+---------+----------+--------------+
only showing top 20 rows
```

```
df.show(20, truncate=False)
```

```
+---------+---------+----------------------------------+--------+--------------+---------+----------+--------------+
|InvoiceNo|StockCode|Description                       |Quantity|InvoiceDate   |UnitPrice|CustomerID|Country       |
+---------+---------+----------------------------------+--------+--------------+---------+----------+--------------+
|536365   |85123A   |WHITE HANGING HEART T-LIGHT HOLDER|6       |12-01-2010 08:26|2.55     |17850     |United Kingdom|
|536365   |71053    |WHITE METAL LANTERN               |6       |12-01-2010 08:26|3.39     |17850     |United Kingdom|
|536365   |84406B   |CREAM CUPID HEARTS COAT HANGER    |8       |12-01-2010 08:26|2.75     |17850     |United Kingdom|
|536365   |84029G   |KNITTED UNION FLAG HOT WATER BOTTLE|6       |12-01-2010 08:26|3.39     |17850     |United Kingdom|
|536365   |84029E   |RED WOOLLY HOTTIE WHITE HEART.    |6       |12-01-2010 08:26|3.39     |17850     |United Kingdom|
|536365   |22752    |SET 7 BABUSHKA NESTING BOXES      |2       |12-01-2010 08:26|7.65     |17850     |United Kingdom|
|536365   |21730    |GLASS STAR FROSTED T-LIGHT HOLDER |6       |12-01-2010 08:26|4.25     |17850     |United Kingdom|
|536366   |22633    |HAND WARMER UNION JACK            |6       |12-01-2010 08:28|1.85     |17850     |United Kingdom|
|536366   |22632    |HAND WARMER RED POLKA DOT         |6       |12-01-2010 08:28|1.85     |17850     |United Kingdom|
|536367   |84879    |ASSORTED COLOUR BIRD ORNAMENT     |32      |12-01-2010 08:34|1.69     |13047     |United Kingdom|
|536367   |22745    |POPPY'S PLAYHOUSE BEDROOM         |6       |12-01-2010 08:34|2.1      |13047     |United Kingdom|
|536367   |22748    |POPPY'S PLAYHOUSE KITCHEN         |6       |12-01-2010 08:34|2.1      |13047     |United Kingdom|
|536367   |22749    |FELTCRAFT PRINCESS CHARLOTTE DOLL |8       |12-01-2010 08:34|3.75     |13047     |United Kingdom|
|536367   |22310    |IVORY KNITTED MUG COSY            |6       |12-01-2010 08:34|1.65     |13047     |United Kingdom|
|536367   |84969    |BOX OF 6 ASSORTED COLOUR TEASPOONS|6       |12-01-2010 08:34|4.25     |13047     |United Kingdom|
|536367   |22623    |BOX OF VINTAGE JIGSAW BLOCKS      |3       |12-01-2010 08:34|4.95     |13047     |United Kingdom|
|536367   |22622    |BOX OF VINTAGE ALPHABET BLOCKS    |2       |12-01-2010 08:34|9.95     |13047     |United Kingdom|
|536367   |21754    |HOME BUILDING BLOCK WORD          |3       |12-01-2010 08:34|5.95     |13047     |United Kingdom|
|536367   |21755    |LOVE BUILDING BLOCK WORD          |3       |12-01-2010 08:34|5.95     |13047     |United Kingdom|
|536367   |21777    |RECIPE BOX WITH METAL HEART       |4       |12-01-2010 08:34|7.95     |13047     |United Kingdom|
+---------+---------+----------------------------------+--------+--------------+---------+----------+--------------+
only showing top 20 rows
```

```
df.printSchema()
```

```
root
 |-- InvoiceNo: string (nullable = true)
 |-- StockCode: string (nullable = true)
 |-- Description: string (nullable = true)
 |-- Quantity: integer (nullable = true)
 |-- InvoiceDate: string (nullable = true)
 |-- UnitPrice: double (nullable = true)
 |-- CustomerID: integer (nullable = true)
 |-- Country: string (nullable = true)
```

```
df.dtypes
```

```
[('InvoiceNo', 'string'),
 ('StockCode', 'string'),
 ('Description', 'string'),
 ('Quantity', 'int'),
 ('InvoiceDate', 'string'),
 ('UnitPrice', 'double'),
 ('CustomerID', 'int'),
 ('Country', 'string')]
```

```
print("Number of records:", df.count())
```

```
Number of records: 541909
```

[9]:
```
from pyspark.sql.functions import col, sum

df.select([
    sum(col(c).isNull().cast("int")).alias(c)
    for c in df.columns
]).show()
```

```
+---------+---------+-----------+--------+-----------+---------+----------+-------+
|InvoiceNo|StockCode|Description|Quantity|InvoiceDate|UnitPrice|CustomerID|Country|
+---------+---------+-----------+--------+-----------+---------+----------+-------+
|        0|        0|       1454|       0|          0|        0|    135080|      0|
+---------+---------+-----------+--------+-----------+---------+----------+-------+
```

[10]:
```
df = df.filter(~col("InvoiceNo").startswith("C"))
```

[11]:
```
df = df.filter(col("Quantity") > 0)
```

[12]:
```
df = df.filter(col("UnitPrice") > 0)
```

[13]:
```
df.show(10)
```

```
+---------+---------+--------------------+--------+--------------+---------+----------+--------------+
|InvoiceNo|StockCode|         Description|Quantity|   InvoiceDate|UnitPrice|CustomerID|       Country|
+---------+---------+--------------------+--------+--------------+---------+----------+--------------+
|   536365|   85123A|WHITE HANGING HEA...|       6|12-01-2010 08:26|     2.55|     17850|United Kingdom|
|   536365|    71053| WHITE METAL LANTERN|       6|12-01-2010 08:26|     3.39|     17850|United Kingdom|
|   536365|   84406B|CREAM CUPID HEART...|       8|12-01-2010 08:26|     2.75|     17850|United Kingdom|
|   536365|   84029G|KNITTED UNION FLA...|       6|12-01-2010 08:26|     3.39|     17850|United Kingdom|
|   536365|   84029E|RED WOOLLY HOTTIE...|       6|12-01-2010 08:26|     3.39|     17850|United Kingdom|
|   536365|    22752|SET 7 BABUSHKA NE...|       2|12-01-2010 08:26|     7.65|     17850|United Kingdom|
|   536365|    21730|GLASS STAR FROSTE...|       6|12-01-2010 08:26|     4.25|     17850|United Kingdom|
|   536366|    22633|HAND WARMER UNION...|       6|12-01-2010 08:28|     1.85|     17850|United Kingdom|
|   536366|    22632|HAND WARMER RED P...|       6|12-01-2010 08:28|     1.85|     17850|United Kingdom|
|   536367|    84879|ASSORTED COLOUR B...|      32|12-01-2010 08:34|     1.69|     13047|United Kingdom|
+---------+---------+--------------------+--------+--------------+---------+----------+--------------+
only showing top 10 rows
```

[14]:
```
df = df.withColumn(
    "Revenue",
    col("Quantity") * col("UnitPrice")
)
```

[15]:
```
df.select(
    "InvoiceNo",
    "Description",
    "Quantity",
    "UnitPrice",
    "Revenue"
).show(10, truncate=False)
```

```
+---------+----------------------------------+--------+---------+------------------+
|InvoiceNo|Description                       |Quantity|UnitPrice|Revenue           |
+---------+----------------------------------+--------+---------+------------------+
|536365   |WHITE HANGING HEART T-LIGHT HOLDER|6       |2.55     |15.299999999999999|
|536365   |WHITE METAL LANTERN               |6       |3.39     |20.34             |
|536365   |CREAM CUPID HEARTS COAT HANGER    |8       |2.75     |22.0              |
|536365   |KNITTED UNION FLAG HOT WATER BOTTLE|6       |3.39     |20.34             |
|536365   |RED WOOLLY HOTTIE WHITE HEART.    |6       |3.39     |20.34             |
|536365   |SET 7 BABUSHKA NESTING BOXES      |2       |7.65     |15.3              |
|536365   |GLASS STAR FROSTED T-LIGHT HOLDER |6       |4.25     |25.5              |
|536366   |HAND WARMER UNION JACK            |6       |1.85     |11.100000000000001|
|536366   |HAND WARMER RED POLKA DOT         |6       |1.85     |11.100000000000001|
|536367   |ASSORTED COLOUR BIRD ORNAMENT     |32      |1.69     |54.08             |
+---------+----------------------------------+--------+---------+------------------+
only showing top 10 rows
```

[16]:
```
df.select(
    sum("Revenue").alias("Total_Revenue")
).show()
```

```
+------------------+
|     Total_Revenue|
+------------------+
|1.0666845144000054E7|
+------------------+
```

[17]:
```
df.select(
    sum("Quantity").alias("Total_Quantity_Sold")
).show()
```

```
+------------------+
|Total_Quantity_Sold|
+------------------+
|           5588376|
+------------------+
```

[18]:
```
total_orders = df.select("InvoiceNo").distinct().count()

print("Total Orders:", total_orders)
```

```
Total Orders: 19960
```

[19]:
```
top_products = df.groupBy("Description") \
    .agg(
        sum("Quantity").alias("Total_Quantity")
    ) \
    .orderBy(desc("Total_Quantity"))

top_products.show(10, truncate=False)
```

```
+---------------------------------+--------------+
|Description                      |Total_Quantity|
+---------------------------------+--------------+
|PAPER CRAFT , LITTLE BIRDIE      |80995         |
|MEDIUM CERAMIC TOP STORAGE JAR   |78033         |
|WORLD WAR 2 GLIDERS ASSTD DESIGNS|55047         |
|JUMBO BAG RED RETROSPOT          |48474         |
|WHITE HANGING HEART T-LIGHT HOLDER|37891        |
|POPCORN HOLDER                   |36761         |
|ASSORTED COLOUR BIRD ORNAMENT    |36461         |
|PACK OF 72 RETROSPOT CAKE CASES  |36419         |
|RABBIT NIGHT LIGHT               |30788         |
|MINI PAINT SET VINTAGE           |26633         |
+---------------------------------+--------------+
only showing top 10 rows
```

[20]:
```
top_revenue_products = df.groupBy("Description") \
    .agg(
        sum("Revenue").alias("Total_Revenue")
    ) \
    .orderBy(desc("Total_Revenue"))

top_revenue_products.show(10, truncate=False)
```

```
+---------------------------------+------------------+
|Description                      |Total_Revenue     |
+---------------------------------+------------------+
|DOTCOM POSTAGE                   |206248.77         |
|REGENCY CAKESTAND 3 TIER         |174484.74         |
|PAPER CRAFT , LITTLE BIRDIE      |168469.6          |
|WHITE HANGING HEART T-LIGHT HOLDER|106292.77000000005|
|PARTY BUNTING                    |99504.32999999997 |
|JUMBO BAG RED RETROSPOT          |94340.05000000002 |
|MEDIUM CERAMIC TOP STORAGE JAR   |81700.92          |
|Manual                           |78112.82          |
|POSTAGE                          |78101.87999999999 |
|RABBIT NIGHT LIGHT               |66964.90999999986 |
+---------------------------------+------------------+
only showing top 10 rows
```

[21]:
```
country_revenue = df.groupBy("Country") \
    .agg(
        sum("Revenue").alias("Total_Revenue")
    ) \
    .orderBy(desc("Total_Revenue"))

country_revenue.show(20, truncate=False)
```

```
+--------------+------------------+
|Country       |Total_Revenue     |
+--------------+------------------+
|United Kingdom|9025222.084000124 |
|Netherlands   |285446.33999999997|
|EIRE          |283453.96000000037|
|Germany       |228867.1399999999 |
|France        |209715.11000000004|
|Australia     |138521.3099999997 |
|Spain         |61577.11          |
|Switzerland   |57009.89999999998 |
|Belgium       |41196.340000000004|
|Sweden        |38378.33          |
|Japan         |37416.37          |
|Norway        |36165.44          |
|Portugal      |33747.1           |
|Finland       |22546.08          |
|Singapore     |21279.29          |
|Channel Islands|20450.440000000005|
|Denmark       |18955.339999999997|
|Italy         |17483.239999999994|
|Hong Kong     |15691.799999999994|
|Cyprus        |13590.380000000001|
+--------------+------------------+
only showing top 20 rows
```

[22]:
```
top_3_countries = df.groupBy("Country") \
    .agg(
        sum("Revenue").alias("Total_Revenue")
    ) \
    .orderBy(desc("Total_Revenue")) \
    .limit(3)

top_3_countries.show()
```

```
+--------------+------------------+
|       Country|     Total_Revenue|
+--------------+------------------+
|United Kingdom| 9025222.084000124|
|   Netherlands|285446.33999999997|
|          EIRE|283453.96000000037|
+--------------+------------------+
```

[23]:
```
invoice_revenue = df.groupBy("InvoiceNo") \
    .agg(
        sum("Revenue").alias("Order_Revenue")
    )
```

[24]:
```
invoice_revenue.select(
    avg("Order_Revenue").alias("Average_Revenue_Per_Order")
).show()
```

```
+-------------------------+
|Average_Revenue_Per_Order|
+-------------------------+
|        534.4030332665335|
+-------------------------+
```

[25]:
```
least_product = df.groupBy("Description") \
    .agg(
        sum("Revenue").alias("Total_Revenue")
    ) \
    .orderBy(asc("Total_Revenue"))

least_product.show(10, truncate=False)
```

```
+-----------------------------------+-------------+
|Description                        |Total_Revenue|
+-----------------------------------+-------------+
|PADS TO MATCH ALL CUSHIONS         |0.003        |
|HEN HOUSE W CHICK IN NEST          |0.42         |
|SET 12 COLOURING PENCILS DOILEY    |0.65         |
|VINTAGE BLUE TINSEL REEL           |0.84         |
|PINK CRYSTAL GUITAR PHONE CHARM    |0.85         |
|HAPPY BIRTHDAY CARD TEDDY/CAKE     |0.95         |
|CAT WITH SUNGLASSES BLANK CARD     |0.95         |
|60 GOLD AND SILVER FAIRY CAKE CASES|1.1          |
|DUSTY PINK CHRISTMAS TREE 30CM     |1.25         |
|FUNKY MONKEY MUG                   |1.25         |
+-----------------------------------+-------------+
only showing top 10 rows
```

[26]:
```
df = df.withColumn(
    "Discount",
    col("Revenue") * 0.10
)
```

[27]:
```
df = df.withColumn(
    "Final_Revenue",
    col("Revenue") - col("Discount")
)
```

[28]:
```
df.select(
    "InvoiceNo",
    "Description",
    "Revenue",
    "Discount",
    "Final_Revenue"
).show(10, truncate=False)
```

```
+---------+-----------------------------------+------------------+------------------+------------------+
|InvoiceNo|Description                        |Revenue           |Discount          |Final_Revenue     |
+---------+-----------------------------------+------------------+------------------+------------------+
|536365   |WHITE HANGING HEART T-LIGHT HOLDER |15.299999999999999|1.53              |13.77             |
|536365   |WHITE METAL LANTERN                |20.34             |2.0340000000000003|18.306            |
|536365   |CREAM CUPID HEARTS COAT HANGER     |22.0              |2.2               |19.8              |
|536365   |KNITTED UNION FLAG HOT WATER BOTTLE|20.34             |2.0340000000000003|18.306            |
|536365   |RED WOOLLY HOTTIE WHITE HEART.     |20.34             |2.0340000000000003|18.306            |
|536365   |SET 7 BABUSHKA NESTING BOXES       |15.3              |1.5300000000000002|13.77             |
|536365   |GLASS STAR FROSTED T-LIGHT HOLDER  |25.5              |2.5500000000000003|22.95             |
|536366   |HAND WARMER UNION JACK             |11.100000000000001|1.11              |9.990000000000002 |
|536366   |HAND WARMER RED POLKA DOT          |11.100000000000001|1.11              |9.990000000000002 |
|536367   |ASSORTED COLOUR BIRD ORNAMENT      |54.08             |5.408             |48.672            |
+---------+-----------------------------------+------------------+------------------+------------------+
only showing top 10 rows
```

[29]:
```
df.select(
    sum("Final_Revenue").alias("Revenue_After_Discount")
).show()
```

```
+----------------------+
|Revenue_After_Discount|
+----------------------+
|     9000816.089600103|
+----------------------+
```

[30]:
```
df = df.withColumn(
    "InvoiceDate",
    coalesce(
        expr("try_to_timestamp(InvoiceDate, 'M/d/yyyy H:mm')"),
        expr("try_to_timestamp(InvoiceDate, 'd-M-yyyy H:mm')")
    )
)
```

[31]:
```
df.printSchema()
```

```
root
 |-- InvoiceNo: string (nullable = true)
 |-- StockCode: string (nullable = true)
 |-- Description: string (nullable = true)
 |-- Quantity: integer (nullable = true)
 |-- InvoiceDate: timestamp (nullable = true)
 |-- UnitPrice: double (nullable = true)
 |-- CustomerID: integer (nullable = true)
 |-- Country: string (nullable = true)
 |-- Revenue: double (nullable = true)
 |-- Discount: double (nullable = true)
 |-- Final_Revenue: double (nullable = true)
```

[33]:
```
monthly_revenue = df.groupBy(
    date_format("InvoiceDate", "yyyy-MM").alias("Month")
).agg(
    round(sum("Revenue"), 2).alias("Monthly_Revenue")
).orderBy("Month")

monthly_revenue.show(20, truncate=False)
```

```
+-------+---------------+
|Month  |Monthly_Revenue|
+-------+---------------+
|2010-01|558960.79      |
|2010-02|477748.38      |
|2010-03|468943.71      |
|2010-05|317749.95      |
|2010-06|548830.46      |
|2010-07|99618.2        |
|2010-08|453389.98      |
|2010-09|535586.18      |
|2010-10|591182.92      |
|2010-12|3257710.57     |
|2011-01|702201.27      |
|2011-02|576737.19      |
|2011-03|758658.51      |
|2011-04|707968.04      |
|2011-05|836693.54      |
|2011-06|788824.86      |
|2011-07|986494.27      |
|2011-08|755231.15      |
|2011-09|1305462.47     |
|2011-10|1029973.78     |
+-------+---------------+
only showing top 20 rows
```

[34]:
```
df.show(20, truncate=False)
```

```
+---------+---------+----------------------------------+--------+-------------------+---------+----------+--------------+-----------------+--------
---------+-----------------+
|InvoiceNo|StockCode|Description                       |Quantity|InvoiceDate        |UnitPrice|CustomerID|Country       |Revenue          |Discount
|Final_Revenue    |
+---------+---------+----------------------------------+--------+-------------------+---------+----------+--------------+-----------------+--------
---------+-----------------+
|536365   |85123A   |WHITE HANGING HEART T-LIGHT HOLDER|6       |2010-01-12 08:26:00|2.55     |17850     |United Kingdom|15.299999999999999|1.53
|13.77            |
|536365   |71053    |WHITE METAL LANTERN               |6       |2010-01-12 08:26:00|3.39     |17850     |United Kingdom|20.34            |2.034000
0000000003|18.306           |
|536365   |84406B   |CREAM CUPID HEARTS COAT HANGER    |8       |2010-01-12 08:26:00|2.75     |17850     |United Kingdom|22.0             |2.2
|19.8             |
|536365   |84029G   |KNITTED UNION FLAG HOT WATER BOTTLE|6       |2010-01-12 08:26:00|3.39     |17850     |United Kingdom|20.34            |2.034000
0000000003|18.306           |
|536365   |84029E   |RED WOOLLY HOTTIE WHITE HEART.    |6       |2010-01-12 08:26:00|3.39     |17850     |United Kingdom|20.34            |2.034000
0000000003|18.306           |
|536365   |22752    |SET 7 BABUSHKA NESTING BOXES      |2       |2010-01-12 08:26:00|7.65     |17850     |United Kingdom|15.3             |1.530000
0000000002|13.77            |
|536365   |21730    |GLASS STAR FROSTED T-LIGHT HOLDER |6       |2010-01-12 08:26:00|4.25     |17850     |United Kingdom|25.5             |2.550000
0000000003|22.95            |
|536366   |22633    |HAND WARMER UNION JACK            |6       |2010-01-12 08:28:00|1.85     |17850     |United Kingdom|11.100000000000001|1.11
|9.990000000000002|
|536366   |22632    |HAND WARMER RED POLKA DOT         |6       |2010-01-12 08:28:00|1.85     |17850     |United Kingdom|11.100000000000001|1.11
|9.990000000000002|
|536367   |84879    |ASSORTED COLOUR BIRD ORNAMENT     |32      |2010-01-12 08:34:00|1.69     |13047     |United Kingdom|54.08            |5.408
|48.672           |
|536367   |22745    |POPPY'S PLAYHOUSE BEDROOM         |6       |2010-01-12 08:34:00|2.1      |13047     |United Kingdom|12.600000000000001|1.260000
0000000002|11.340000000000002|
|536367   |22748    |POPPY'S PLAYHOUSE KITCHEN         |6       |2010-01-12 08:34:00|2.1      |13047     |United Kingdom|12.600000000000001|1.260000
0000000002|11.340000000000002|
|536367   |22749    |FELTCRAFT PRINCESS CHARLOTTE DOLL |8       |2010-01-12 08:34:00|3.75     |13047     |United Kingdom|30.0             |3.0
|27.0             |
|536367   |21310    |IVORY KNITTED MUG COSY            |6       |2010-01-12 08:34:00|1.65     |13047     |United Kingdom|9.899999999999999|0.989999
9999999999|8.909999999999998|
|536367   |84969    |BOX OF 6 ASSORTED COLOUR TEASPOONS|6       |2010-01-12 08:34:00|4.25     |13047     |United Kingdom|25.5             |2.550000
0000000003|22.95            |
|536367   |22623    |BOX OF VINTAGE JIGSAW BLOCKS      |3       |2010-01-12 08:34:00|4.95     |13047     |United Kingdom|14.850000000000001|1.485000
0000000003|13.365000000000002|
|536367   |22622    |BOX OF VINTAGE ALPHABET BLOCKS    |2       |2010-01-12 08:34:00|9.95     |13047     |United Kingdom|19.9             |1.99
|17.91            |
|536367   |21754    |HOME BUILDING BLOCK WORD          |3       |2010-01-12 08:34:00|5.95     |13047     |United Kingdom|17.85            |1.785000
0000000001|16.065           |
|536367   |21755    |LOVE BUILDING BLOCK WORD          |3       |2010-01-12 08:34:00|5.95     |13047     |United Kingdom|17.85            |1.785000
0000000001|16.065           |
|536367   |21777    |RECIPE BOX WITH METAL HEART       |4       |2010-01-12 08:34:00|7.95     |13047     |United Kingdom|31.8             |3.18
|28.62            |
+---------+---------+----------------------------------+--------+-------------------+---------+----------+--------------+-----------------+--------
---------+-----------------+
only showing top 20 rows
```

# Conclusion

The E-Commerce Sales Analysis using Spark DataFrames practical was successfully completed using PySpark and the UCI Online Retail dataset. The dataset was loaded into a Spark DataFrame and its schema and data types were examined. Missing values were identified and invalid transactions were filtered. A Revenue column was created using Quantity × UnitPrice, followed by product-wise and country-wise revenue analysis. The practical also calculated total quantity sold, total orders, average revenue per order, discount-based revenue and monthly revenue summaries. The analysis demonstrated the use of Spark DataFrame operations such as filtering, withColumn(), groupBy(), aggregation and sorting for large-scale data processing. The practical provided a better understanding of Spark DataFrames, data cleaning, revenue analysis, aggregation, scalable processing and business intelligence reporting.